

Behrooz Vedadian

AI RESEARCHER | DEEP LEARNING THEORY & ARCHITECTURES

✉ vedadian@gmail.com | <https://vedadian.com> | vedadian | [in](#) behrooz-vedadian

Profile

Mathematics enthusiast with a strong interest in the theoretical foundations of deep learning and learning theory. Also an experienced engineer with substantial research history in machine learning and neural architectures.

Education

Ph.D. in Artificial Intelligence **Amirkabir University of Technology** 2013 – 2017

- Supervisors: *Dr. Mohammad Rahmati, Dr. Shahram Khadivi*
- Research direction: *Automatic Machine Translation and graph-based neural modeling*
- Coursework and research completed

M.Sc. in Communication Systems **Amirkabir University of Technology** 2005 – 2008

- Supervisors: *Dr. Hamidreza Amindavar*
- Thesis: *Signal Detection in Nonstationary Environments using Wavelets*

B.Sc. in Electrical Engineering **Sharif University of Technology** 2000 – 2005

- Supervisors: *Dr. Mahdi Nayebi*
- Thesis: *Speaker Identification using Hidden Markov Models*

Research & Professional Experience

Targoman Intelligent Processing Co. **Researcher, Developer & Co-founder** 2015 – 2024

- Neural machine translation model training and deployment.
- PDF Document Layout Analysis.

Projects **Researcher & Developer**

- Yara Group: Defect detection in Dental Xray images using deep learning 2021.
- Faraadid Co.: License plate recognition, Surveillance video analysis and object tracking 2017.

HLT Lab, Amirkabir University of Technology **Researcher** 2013 – 2015

- Statistical machine translation model training and deployment.

Hoda Systems Co. **Researcher & Developer** 2005 – 2013

- Optical Character Recognition and Document Layout Analysis.

Selected Research Artifacts

- Graph signal processing-based neural architecture for sequence modeling (PhD research proposal).
- Technical documentation for large-scale Persian neural MT systems.

Awards

- Ranked 9th in National Ph.D. Entrance Examination (2013).
- National Programming Contest medalist (1997, 1998).

Technical Skills

C++ · Python · PyTorch · TensorFlow · Linear algebra · Optimization